

# CSE428: Image Processing

## Lecture 1: Introduction

June 2, 2022



# Outline

- Signal and Data
- What is Image
- Digital Image
- Computer Representation
- Types of Digital Images
- Image Processing - Examples
- Application of Image Processing

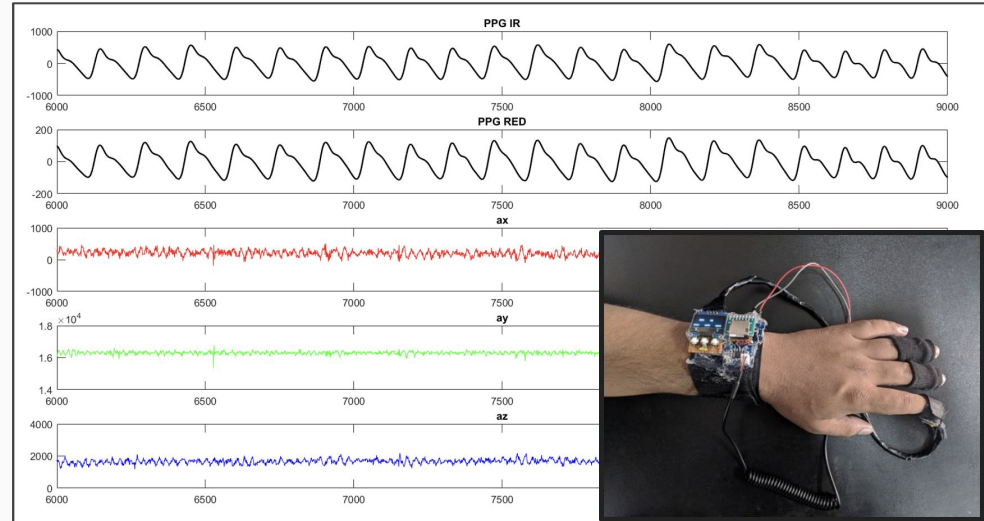
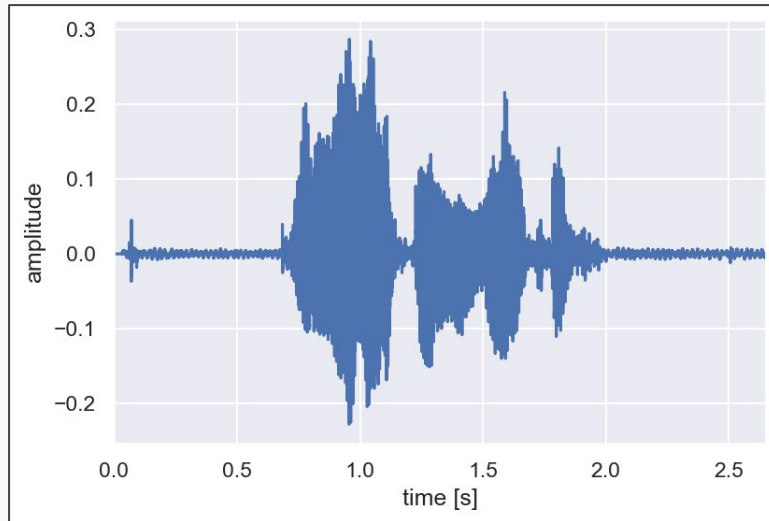
# Signal (Data)

- A **function** containing **information** about some phenomena
- Any physical quantity (temperature, light intensity, position) varying with time (t) or space (x, y, z...)
- $f(t)$ ,  $f(x, y)$ ,  $f(x, t)$



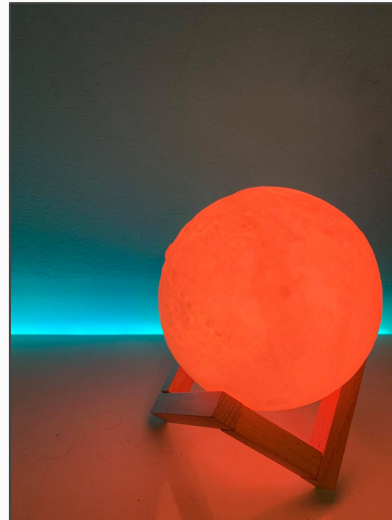
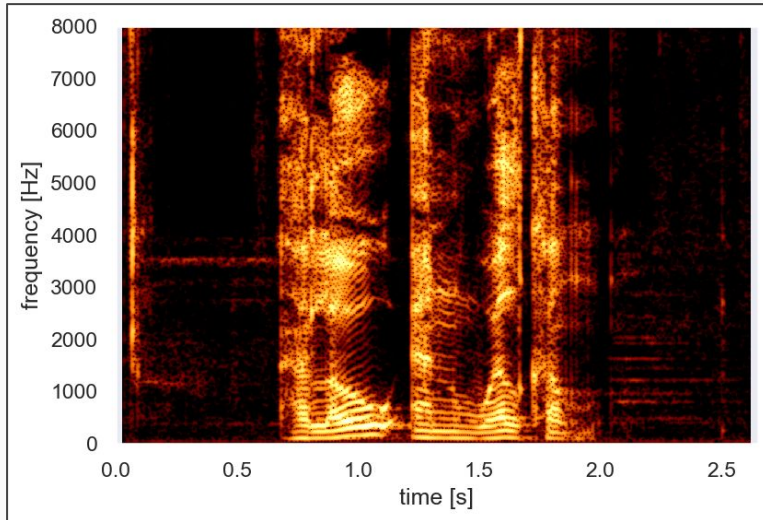
# Examples of Signal - 1D

1D signal -  $f(x)$  or  $f(t)$



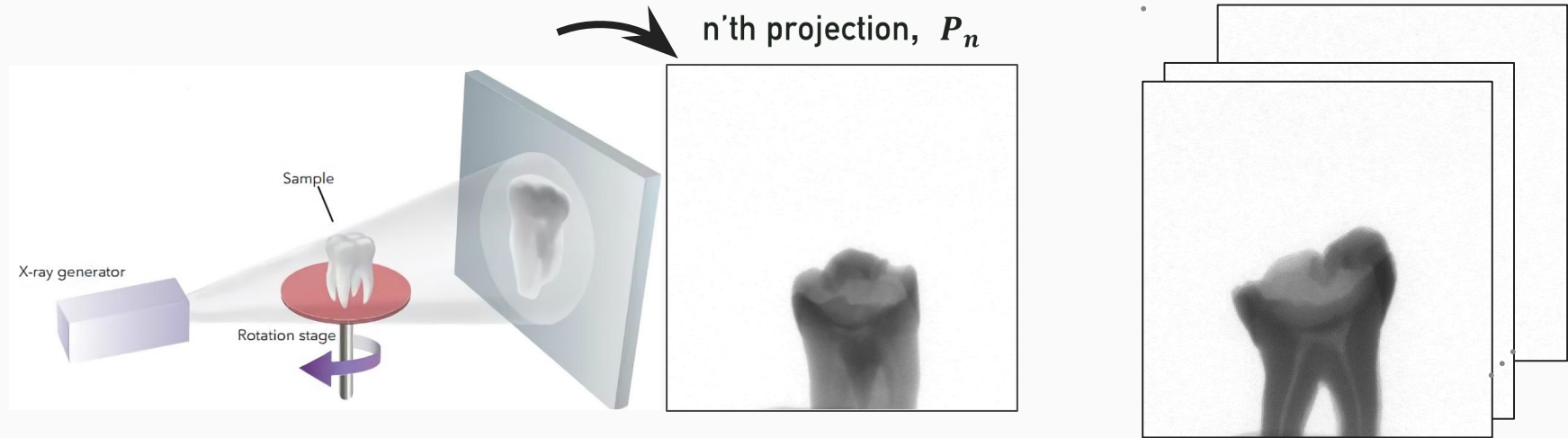
# Examples of Signal - 2D

2D signal -  $f(x, y)$  or  $f(x, t)$



# Examples of Signal - 3D

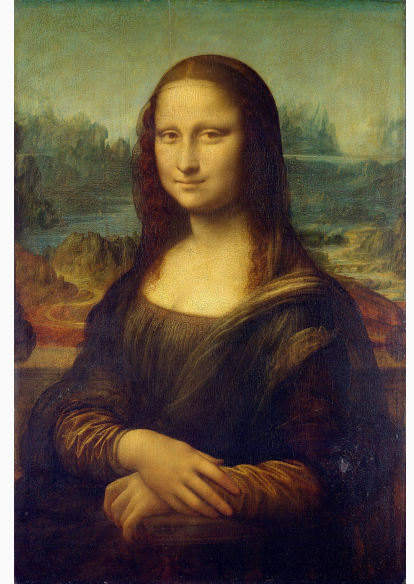
3D signal -  $f(x, y, t)$



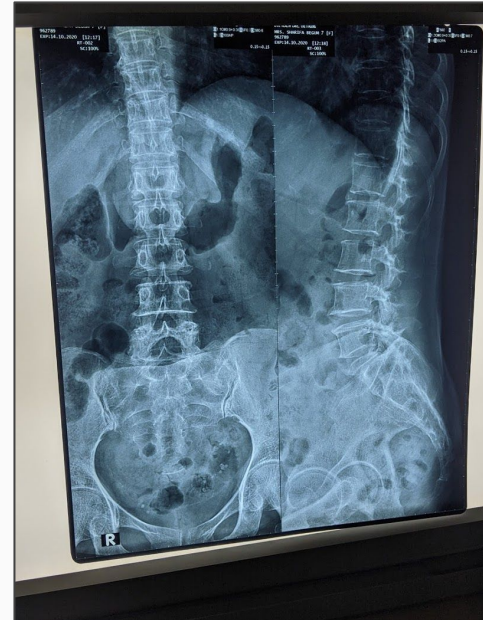
# Image

- A visual representation in the form  $f(x, y)$
- $f \rightarrow$  brightness or color at  $(x, y)$
- $(x, y)$  are called spatial coordinates
- Usually  $f$ ,  $x$ , and  $y$  have continuous values

# Image - Paintings

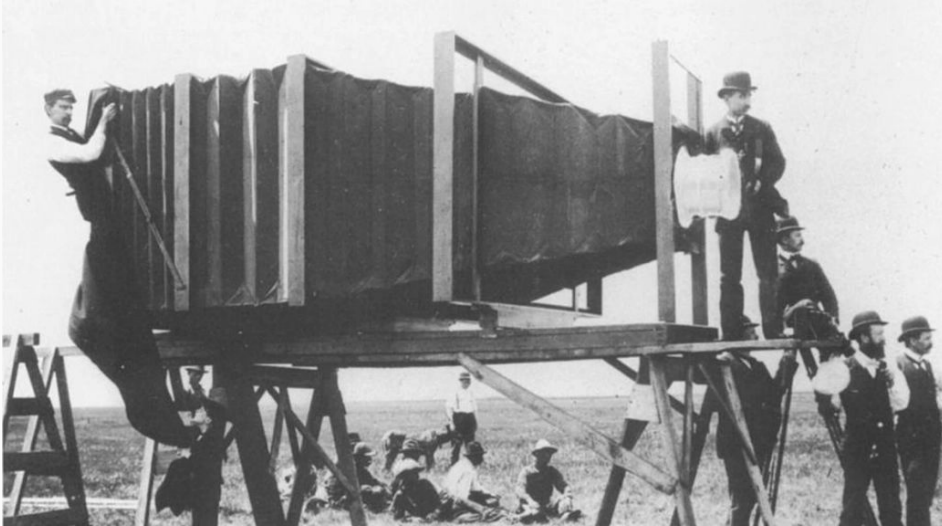


# Image - Medical Images



# Image - “Traditional” Photography

Optical Array → Sensor (Film) → Chemicals → Final Image



# Image - “Traditional” Photography

Optical Array → Sensor (Film) → Chemicals → Final Image



Cannot be stored!  
(Digitally)

# Digital Image

- Discrete (and finite) samples  $f[x, y]$
- Composed of finite number of elements - **pixels**
- $x, y \rightarrow$  Pixel location or index (0, 1, 2, 3, .... ~~1.5, 2.345~~ )
- $f[x, y] \rightarrow$  particular value(s)

400 x 400



200 x 200



100 x 100



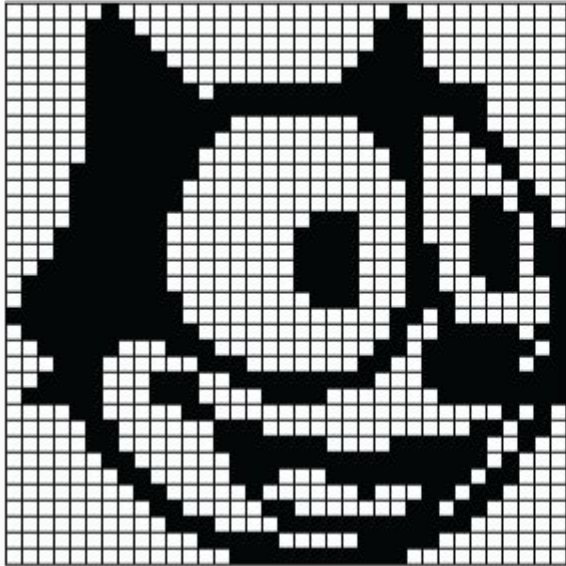
50 x 50



25 x 25

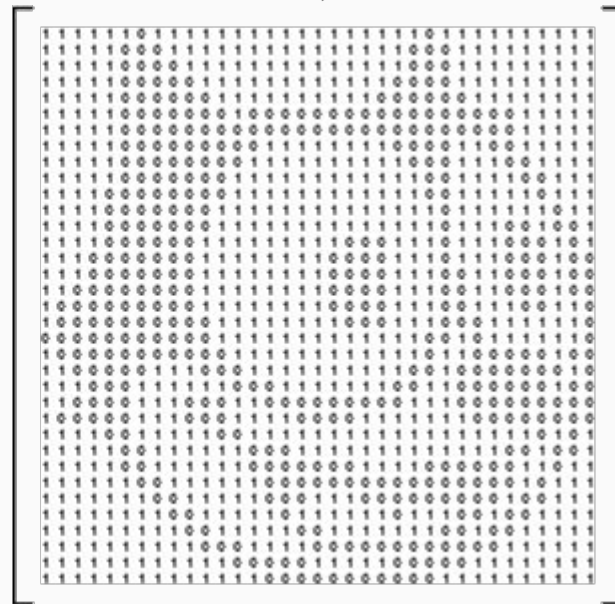
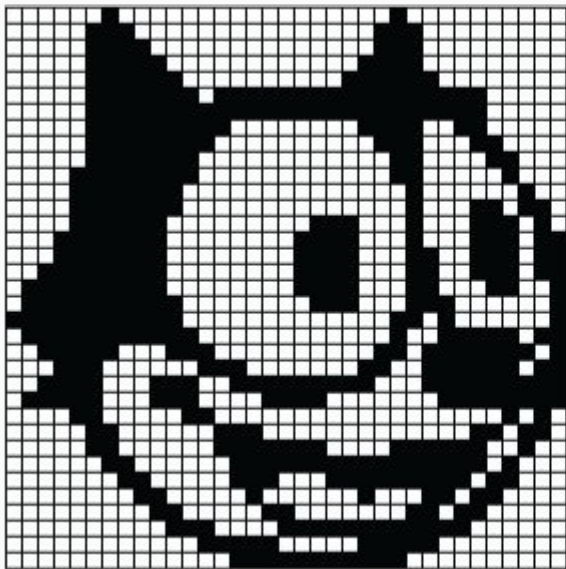


# Storing Digital Image - Black & White



# Storing Digital Image - Black & White

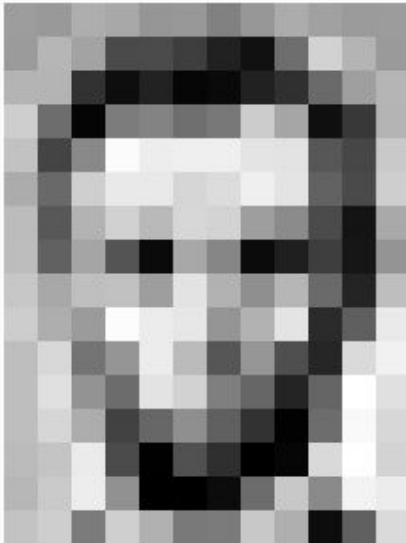
Images are stored as matrix (high -> white, low -> black)



35x35

# Storing Digital Image - Grayscale

Images are stored as matrix (high -> white, low -> black, in-between -> gray)



|     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| 157 | 153 | 174 | 168 | 150 | 152 | 129 | 151 | 172 | 161 | 155 | 156 |
| 155 | 182 | 163 | 74  | 75  | 62  | 83  | 17  | 110 | 210 | 180 | 154 |
| 180 | 180 | 50  | 14  | 34  | 6   | 10  | 83  | 48  | 106 | 159 | 181 |
| 206 | 109 | 5   | 124 | 131 | 111 | 120 | 204 | 166 | 15  | 56  | 180 |
| 194 | 68  | 137 | 251 | 237 | 239 | 239 | 228 | 227 | 87  | 71  | 201 |
| 172 | 106 | 207 | 233 | 233 | 214 | 220 | 239 | 228 | 98  | 74  | 206 |
| 188 | 88  | 179 | 209 | 185 | 215 | 211 | 158 | 139 | 75  | 20  | 169 |
| 189 | 97  | 165 | 84  | 10  | 168 | 134 | 11  | 31  | 62  | 22  | 148 |
| 199 | 168 | 191 | 193 | 158 | 227 | 178 | 143 | 182 | 106 | 36  | 190 |
| 205 | 174 | 155 | 252 | 236 | 231 | 149 | 178 | 228 | 43  | 95  | 234 |
| 190 | 216 | 116 | 149 | 236 | 187 | 86  | 150 | 79  | 38  | 218 | 241 |
| 190 | 224 | 147 | 108 | 227 | 210 | 127 | 102 | 36  | 101 | 255 | 224 |
| 190 | 214 | 173 | 66  | 103 | 143 | 96  | 50  | 2   | 109 | 249 | 215 |
| 187 | 196 | 235 | 75  | 1   | 81  | 47  | 0   | 6   | 217 | 255 | 211 |
| 183 | 202 | 237 | 145 | 0   | 0   | 12  | 108 | 200 | 138 | 243 | 236 |
| 195 | 206 | 123 | 207 | 177 | 121 | 123 | 205 | 175 | 13  | 96  | 218 |

|     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| 157 | 153 | 174 | 168 | 150 | 152 | 129 | 151 | 172 | 161 | 155 | 156 |
| 155 | 182 | 163 | 74  | 75  | 62  | 83  | 17  | 110 | 210 | 180 | 154 |
| 180 | 180 | 50  | 14  | 34  | 6   | 10  | 83  | 48  | 106 | 159 | 181 |
| 206 | 109 | 5   | 124 | 131 | 111 | 120 | 204 | 166 | 15  | 56  | 180 |
| 194 | 68  | 137 | 251 | 237 | 239 | 239 | 228 | 227 | 87  | 71  | 201 |
| 172 | 106 | 207 | 233 | 233 | 214 | 220 | 239 | 228 | 98  | 74  | 206 |
| 188 | 88  | 179 | 209 | 185 | 215 | 211 | 158 | 139 | 75  | 20  | 169 |
| 189 | 97  | 165 | 84  | 10  | 168 | 134 | 11  | 31  | 62  | 22  | 148 |
| 199 | 168 | 191 | 193 | 158 | 227 | 178 | 143 | 182 | 106 | 36  | 190 |
| 205 | 174 | 155 | 252 | 236 | 231 | 149 | 178 | 228 | 43  | 95  | 234 |
| 190 | 216 | 116 | 149 | 236 | 187 | 86  | 150 | 79  | 38  | 218 | 241 |
| 190 | 224 | 147 | 108 | 227 | 210 | 127 | 102 | 36  | 101 | 255 | 224 |
| 190 | 214 | 173 | 66  | 103 | 143 | 96  | 50  | 2   | 109 | 249 | 215 |
| 187 | 196 | 235 | 75  | 1   | 81  | 47  | 0   | 6   | 217 | 255 | 211 |
| 183 | 202 | 237 | 145 | 0   | 0   | 12  | 108 | 200 | 138 | 243 | 236 |
| 195 | 206 | 123 | 207 | 177 | 121 | 123 | 205 | 175 | 13  | 96  | 218 |

# Computer Representation - Grayscale Version 1

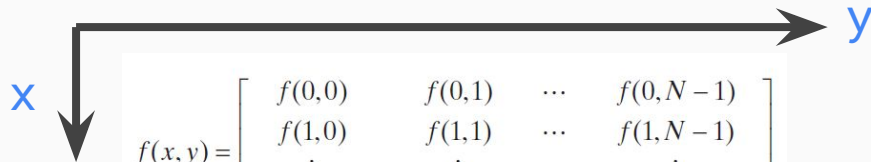
- For an image with M rows and N columns
  - $x = 0, 1, 2, \dots, (N - 1)$  where N is the **width** of the image (also called W)
  - $y = 0, 1, 2, \dots, (M - 1)$  where M is the **height** of the image (also called H)
- Numerical array form  $[f(x, y)]$
- $(i, j)$  th pixel value  $[f(i, j)]$  is the image intensity at point  $(i, j)$
- In Python - `img[y, x]`, `img.shape = (H, W)`

$$f(x, y) = \begin{bmatrix} f(0,0) & f(1,0) & \dots & f(N-1,0) \\ f(0,1) & f(1,1) & \dots & f(N-1,1) \\ \vdots & \vdots & f(i,j) & \vdots \\ f(0,M-1) & f(1,M-1) & \dots & f(N-1,M-1) \end{bmatrix} = \text{img}[y, x] = \begin{bmatrix} \text{im}(0,0) & \text{im}(0,1) & \dots & \text{im}(0,N-1) \\ \text{im}(1,0) & \text{im}(1,1) & \dots & \text{im}(1,N-1) \\ \vdots & \vdots & \text{im}(j,i) & \vdots \\ \text{im}(M-1,0) & \text{im}(M-1,1) & \dots & \text{im}(M-1,N-1) \end{bmatrix}$$

# Computer Representation - Grayscale

## Version 2

- For an image with M rows and N columns
  - $x = 0, 1, 2, \dots, (M - 1)$  where M is the **height** of the image (also called H)
  - $y = 0, 1, 2, \dots, (N - 1)$  where N is the **width** of the image (also called W)
- Numerical array form  $[f(x, y)]$
- $(i, j)$  th pixel value  $[f(i, j)]$  is the image intensity at point  $(i, j)$
- In Python - `img[x, y]`, `img.shape = (H, W)`


$$f(x,y) = \begin{bmatrix} f(0,0) & f(0,1) & \cdots & f(0,N-1) \\ f(1,0) & f(1,1) & \cdots & f(1,N-1) \\ \vdots & \vdots & & \vdots \\ f(M-1,0) & f(M-1,1) & \cdots & f(M-1,N-1) \end{bmatrix}$$

$$\mathbf{A} = \begin{bmatrix} a_{0,0} & a_{0,1} & \cdots & a_{0,N-1} \\ a_{1,0} & a_{1,1} & \cdots & a_{1,N-1} \\ \vdots & \vdots & & \vdots \\ a_{M-1,0} & a_{M-1,1} & \cdots & a_{M-1,N-1} \end{bmatrix}$$

# Storing Digital Image - Color Image

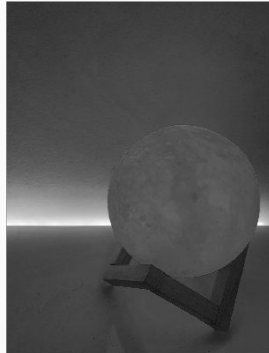
## Version 1

- Color image = 3 grayscale image (RGB model)
- **R** = Red, **G** = Green, **B** = Blue [each grayscale images are called channel]
- Numerical array form  $[f(x, y, n)]$ ,  $n$  = index of channel = 0, 1, 2
- In Python - `img[y, x, n]`, `img.shape = (H, W, #channels)`

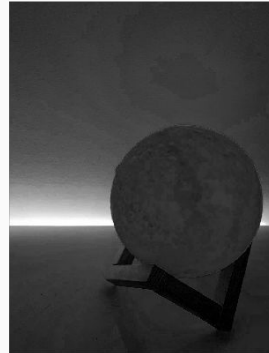
Red



Green



Blue



# Storing Digital Image - Color Image

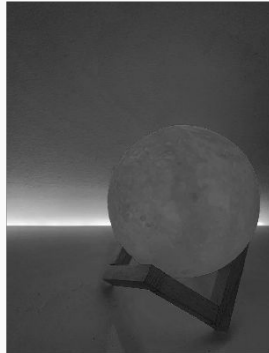
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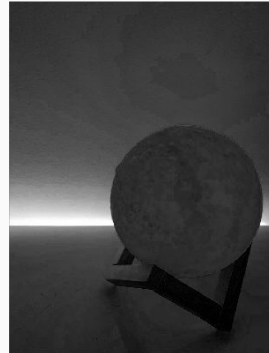
Red



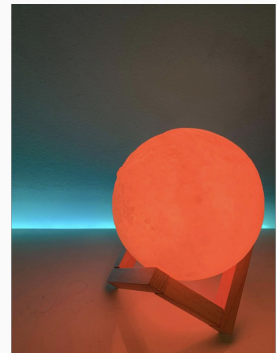
Green



Blue



Combined



# Storing Digital Image - Color Image

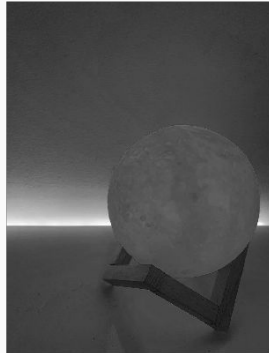
## Version 2

- Color image = 3 grayscale image (RGB model)
- **R** = Red, **G** = Green, **B** = Blue [each grayscale images are called channel]
- Numerical array form  $[f(x, y, n)]$ ,  $n$  = index of channel = 0, 1, 2
- In Python - `img[x, y, n]`, `img.shape = (H, W, #channels)`

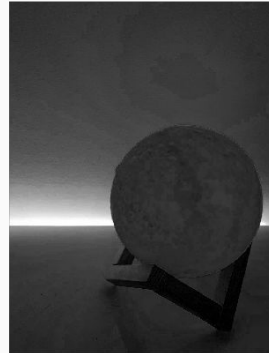
Red



Green



Blue



# Storing Digital Image - Color Image

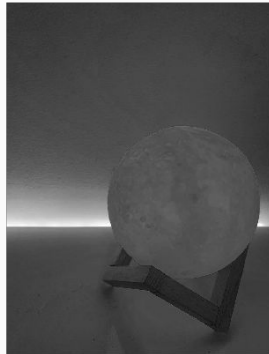
## Version 2

- Color image = 3 grayscale image (RGB model)
- **R** = Red, **G** = Green, **B** = Blue [each grayscale images are called channel]
- Numerical array form  $[f(x, y, n)]$ ,  $n$  = index of channel = 0, 1, 2
- In Python - `img[x, y, n]`, `img.shape = (H, W, #channels)`

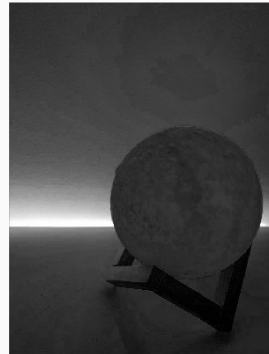
Red



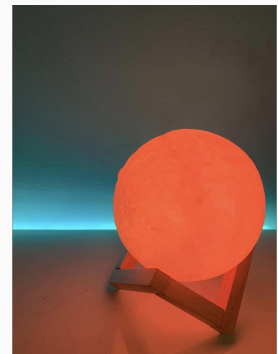
Green



Blue

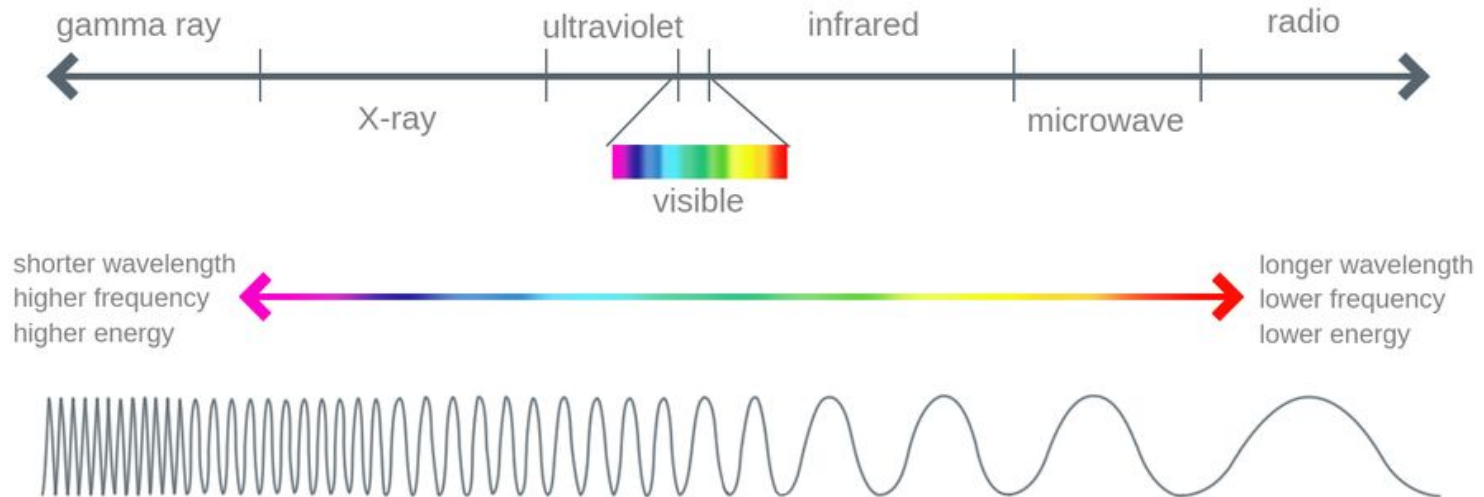


Combined



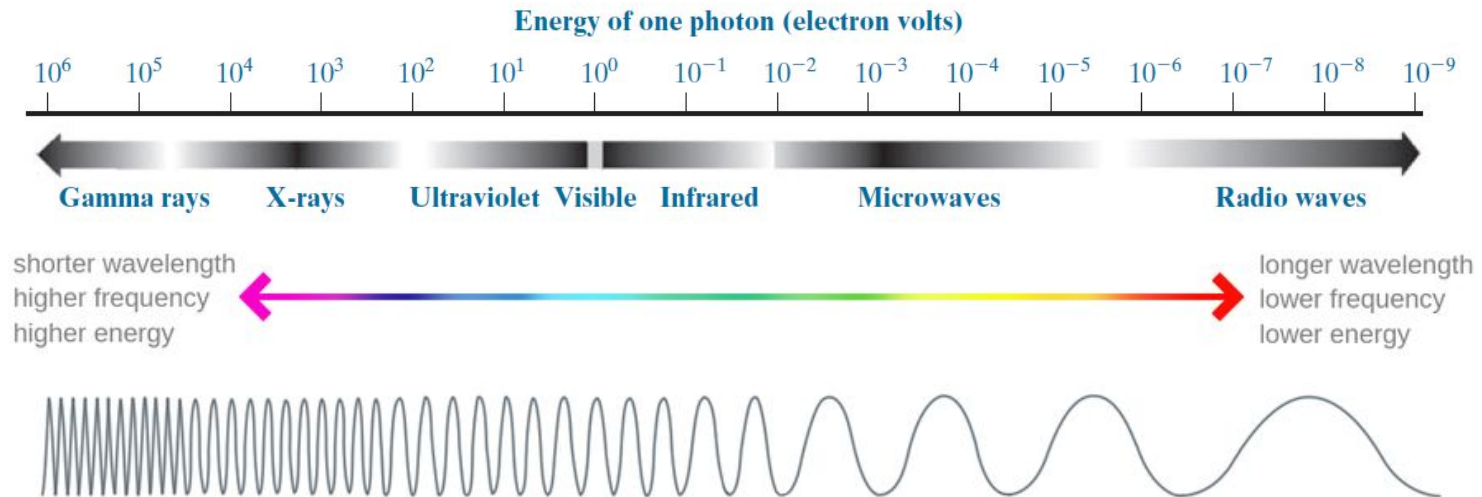
# Classification of Digital Image

According to their source\*



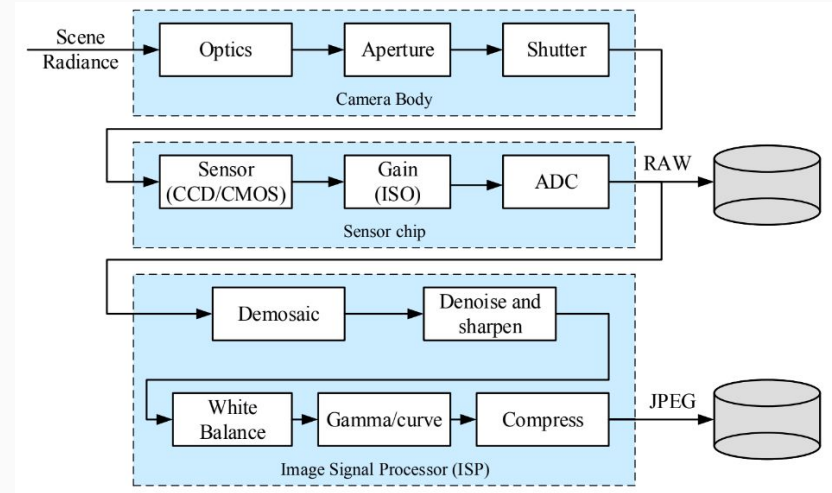
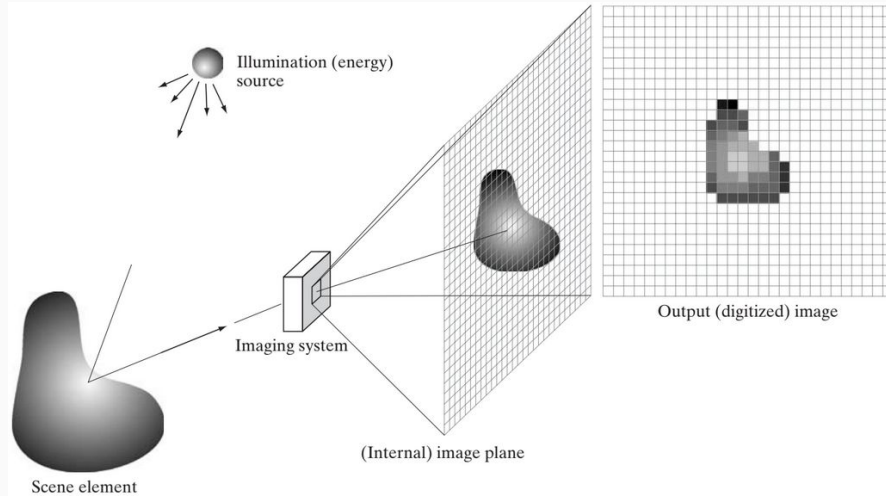
# Classification of Digital Image

## According to their source\*



# Classification of Digital Image

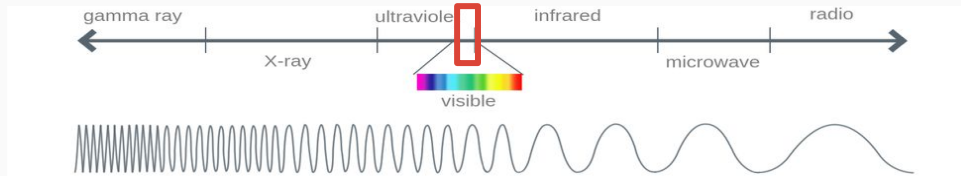
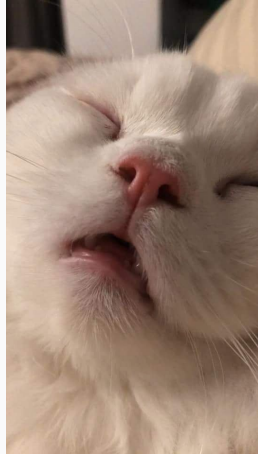
## Reflection Image



Information primarily about object's surface

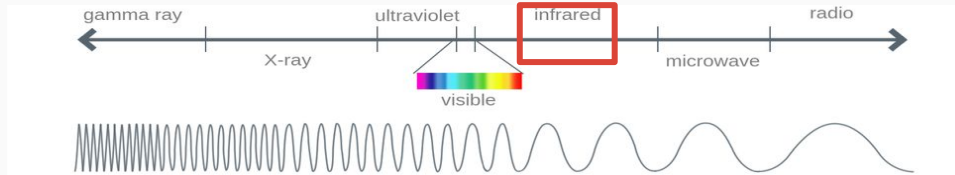
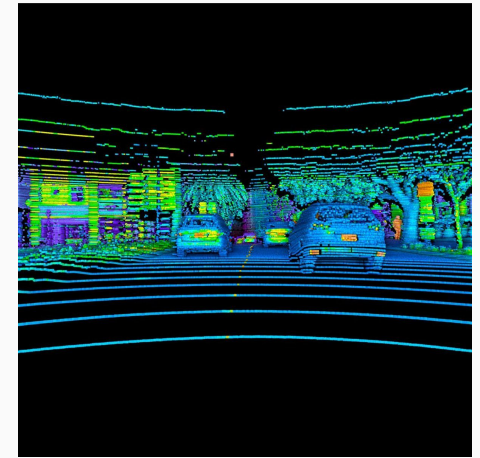
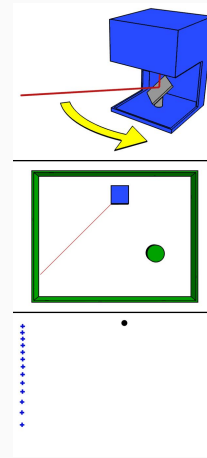
# Classification of Digital Image

## Reflection Image - Visible



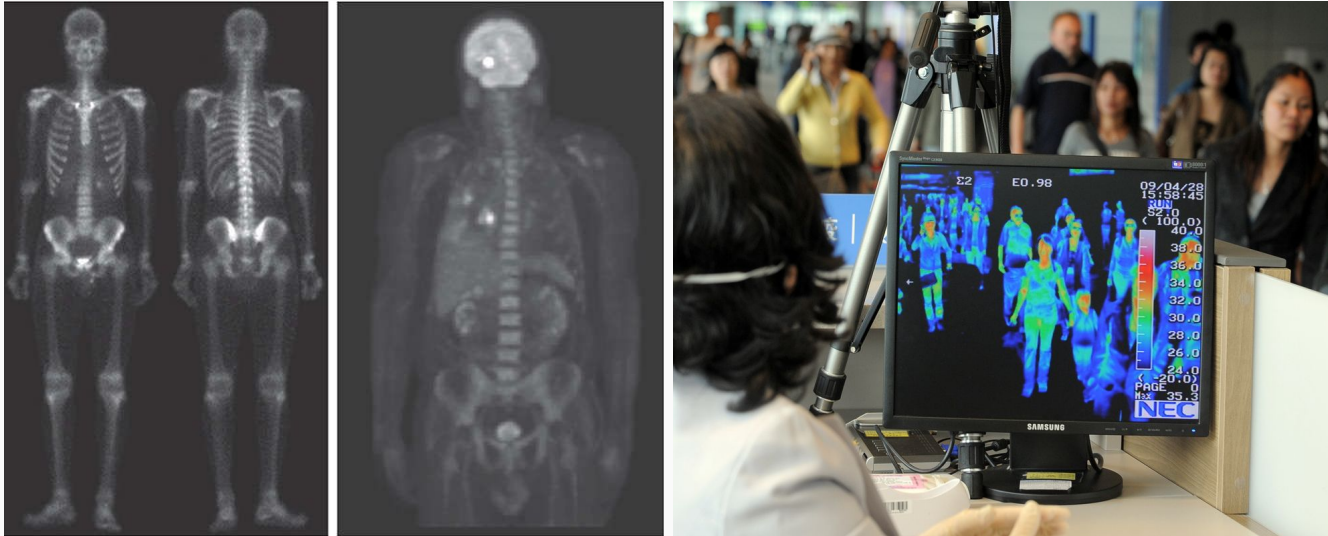
# Classification of Digital Image

## Reflection Image - LiDAR



# Classification of Digital Image

## Emission Image

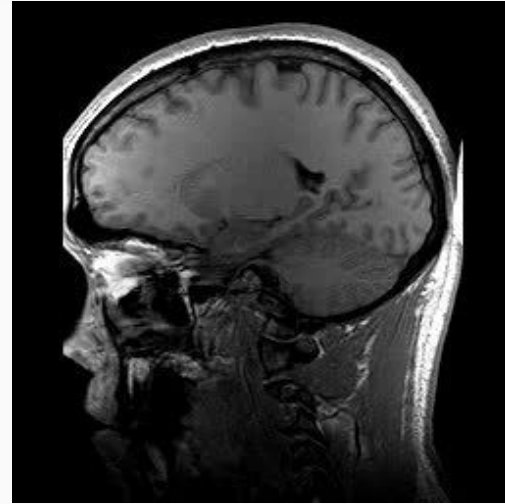


- (a) Gamma ray
- (b) PET Image (gamma)
- (c) Infrared (IR)

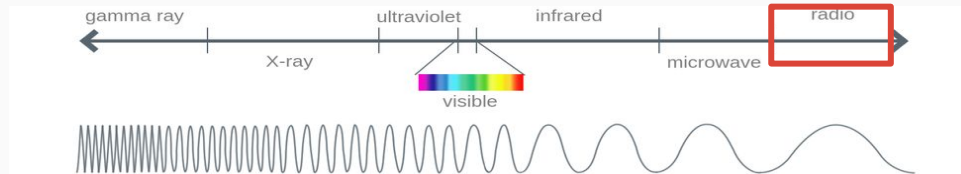
Information primarily about object's Internal

# Classification of Digital Image

## Emission Image

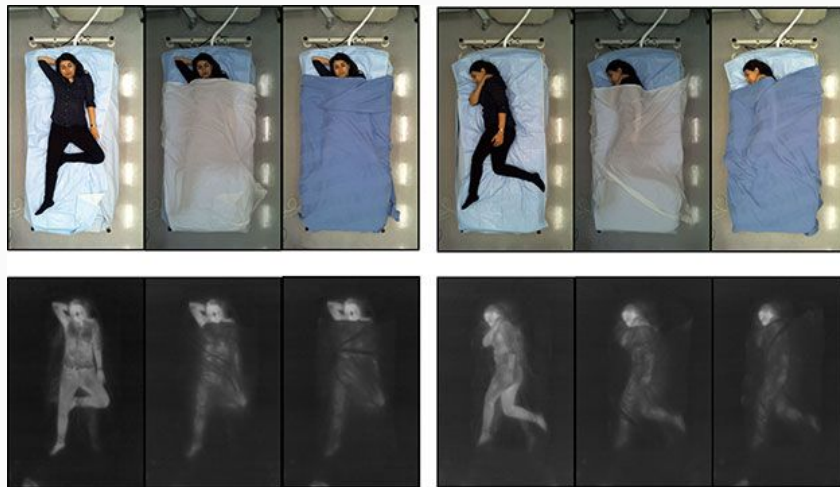


MRI Image

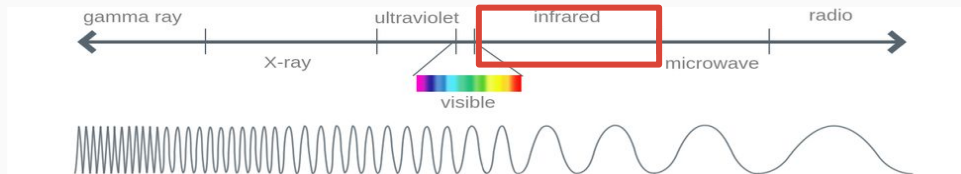


# Classification of Digital Image

## Emission Image

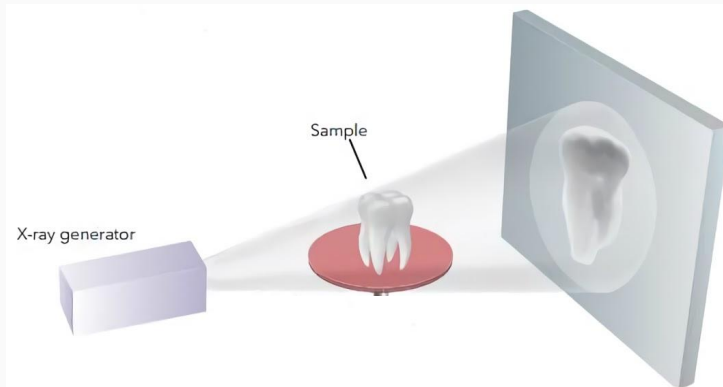


VIP Cup 2021 Challenge  
SLP Pose Estimation

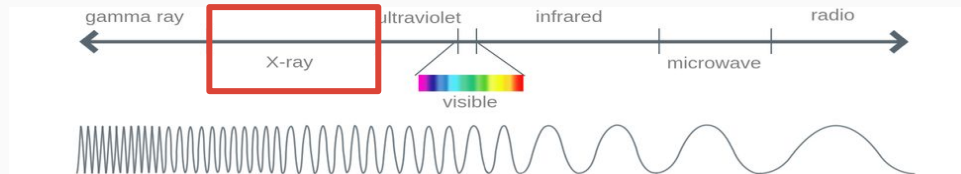


# Classification of Digital Image

## Absorption Image - X Ray



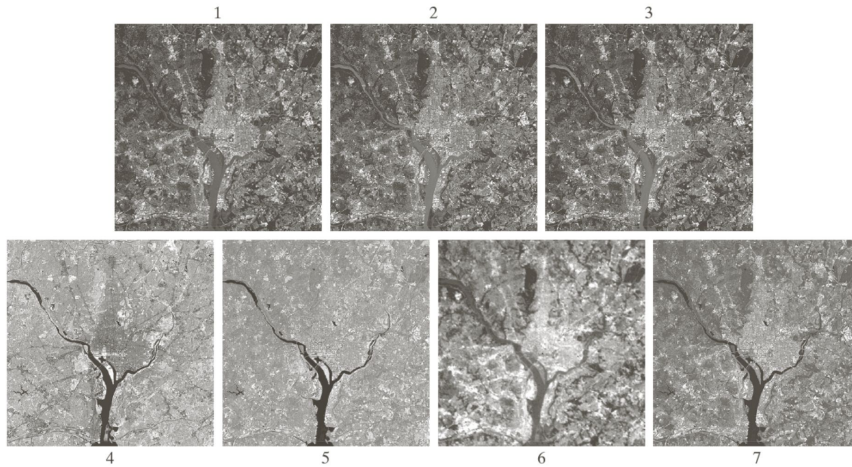
X Ray



Information primarily about object's Internal structure

# Classification of Digital Image

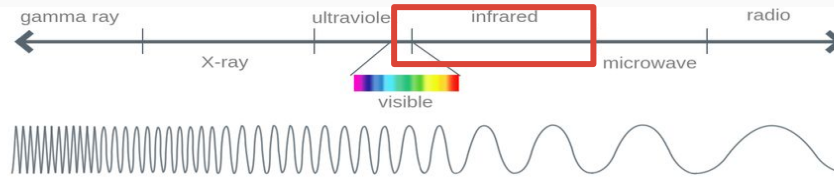
## Multimodal Image - LANDSAT Satellite Image



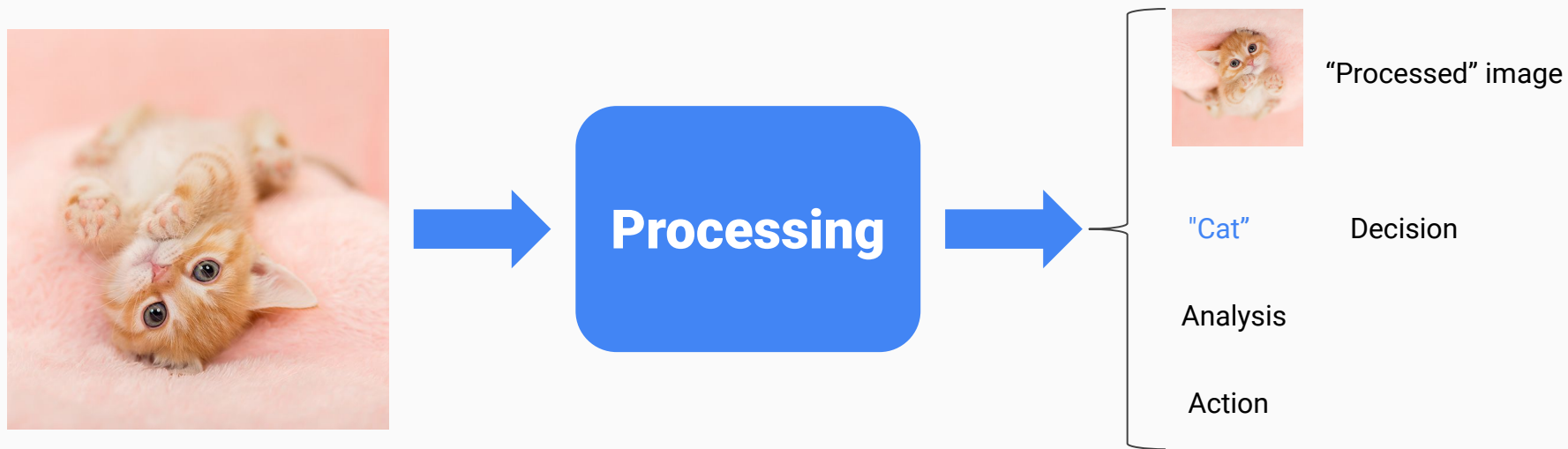
**FIGURE 1.10** LANDSAT satellite images of the Washington, D.C. area. The numbers refer to the thematic bands in Table 1.1. (Images courtesy of NASA.)

| Band No. | Name             | Wavelength ( $\mu\text{m}$ ) | Characteristics and Uses                |
|----------|------------------|------------------------------|-----------------------------------------|
| 1        | Visible blue     | 0.45–0.52                    | Maximum water penetration               |
| 2        | Visible green    | 0.52–0.60                    | Good for measuring plant vigor          |
| 3        | Visible red      | 0.63–0.69                    | Vegetation discrimination               |
| 4        | Near infrared    | 0.76–0.90                    | Biomass and shoreline mapping           |
| 5        | Middle infrared  | 1.55–1.75                    | Moisture content of soil and vegetation |
| 6        | Thermal infrared | 10.4–12.5                    | Soil moisture; thermal mapping          |
| 7        | Middle infrared  | 2.08–2.35                    | Mineral mapping                         |

**TABLE 1.1**  
Thematic bands in NASA's LANDSAT satellite.



# Image Processing



# Image Processing - Examples

## Image Enhancement

Before:



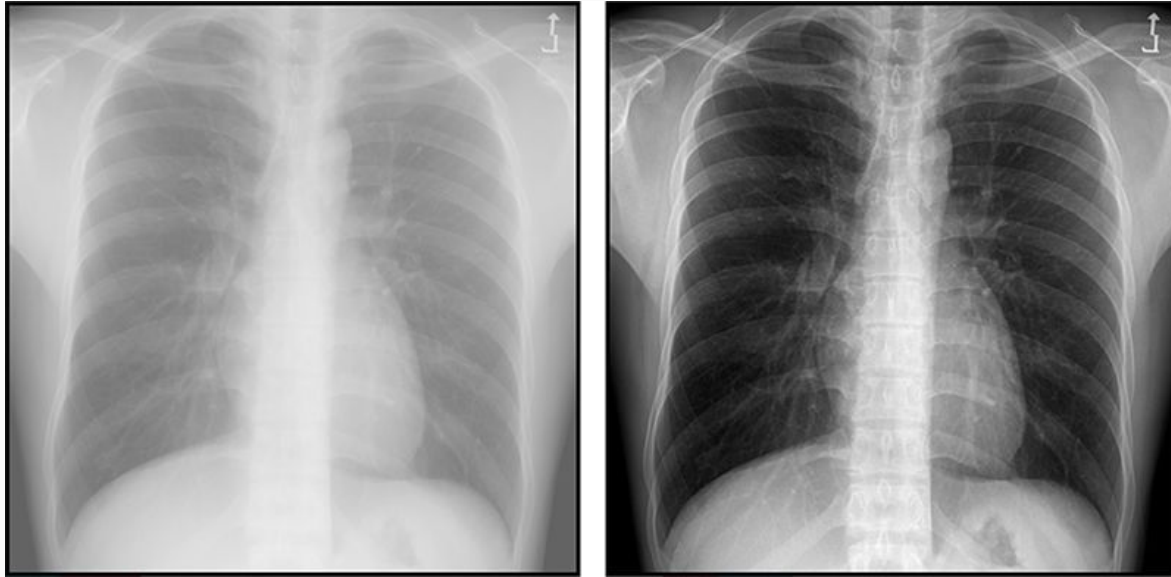
After:



Contrast Limited Adaptive **Histogram Equalization**

# Image Processing - Examples

## Image Enhancement



Intensity transformation, Contrast enhancement

# Image Processing - Examples

## Image Enhancement



RAW



HDR+

HDR Image, Computational Photography

# Image Processing - Examples

## Image Noise Reduction/Removal



Input Frame

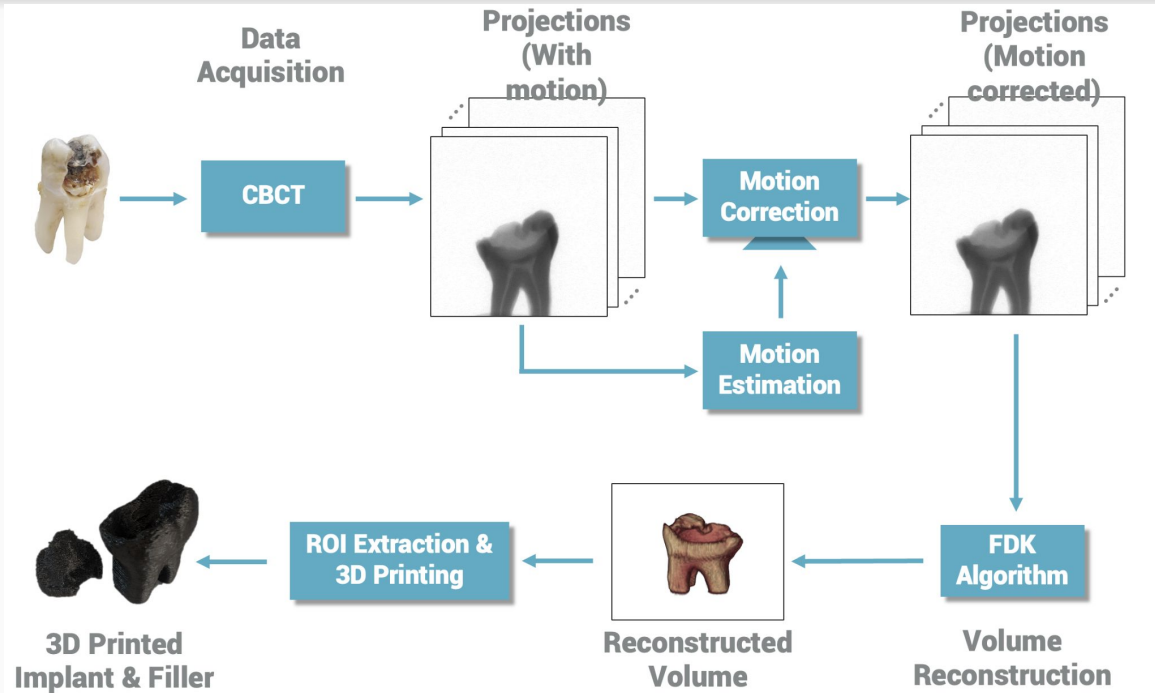


Output Frame

CNN Based de-Raining

# Image Processing - Examples

## Image Reconstruction



# Image Processing - Examples

Image Recognition (Computer Vision)

**Classification**



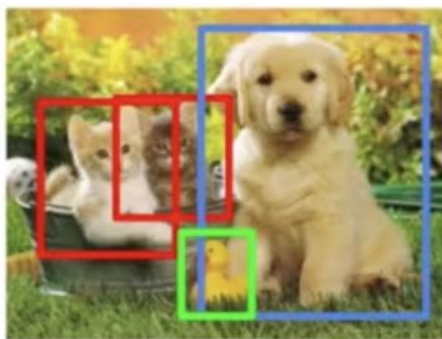
CAT

**Classification  
+ Localization**



CAT

**Object Detection**



CAT, DOG, DUCK

**Instance  
Segmentation**

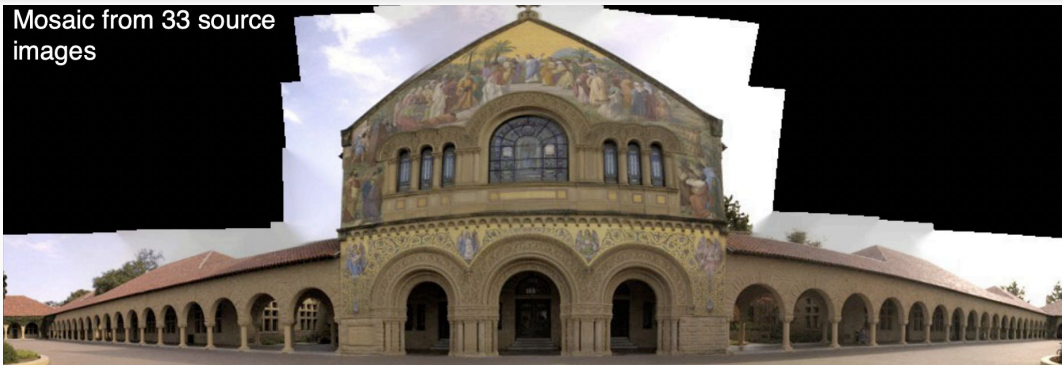


CAT, DOG, DUCK

# Image Processing - Examples

## Image Stitching

Mosaic from 33 source images



Mosaic from 21 source images

# Application of Image Processing

- **Medicine**
  - Medical Imaging, anomaly detection, automatic disease detection
- **Machine Vision**
  - Quality control
  - Measurement of position, orientation, sorting
- **Military**
  - Battlefield awareness
  - Object tracing
- **Autonomous Vehicles**
  - Self-driving cars (Tesla)
  - Self exploration (NASA's Curiosity, Perseverance)

Questions?